Sage FlashPoint

Multi-node lightning localization & wildfire ignition watch on the Sage fleet

Cameras catch the flash. Microphones catch the thunder. Weather decides when to listen.

Samuel Watson · University of Hawaiʻi at Mānoa · Sage Summer Camp 2026 hackathon

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PRESENTATION OVERVIEW

The next ten minutes, in eight stops

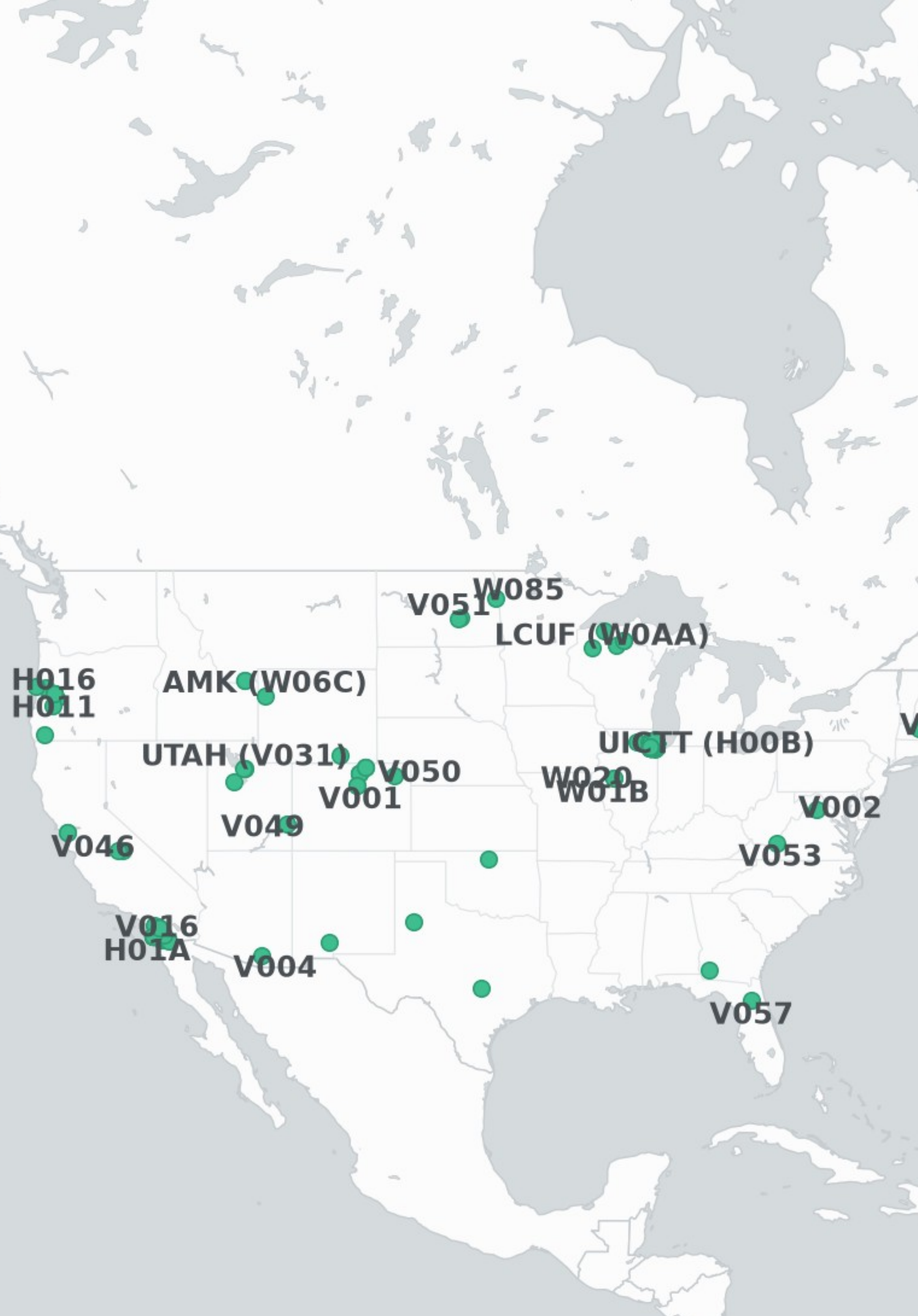
1. **The Sage platform**
   a continent of sensors at the edge
2. **Sage FlashPoint in one picture**
   the flow of state, from quiet skies to a verified alert
3. **Why it matters**
   lightning holdover fires start where nobody is watching
4. **The data**
   five years of fleet archives plus four public feeds
5. **Cases observed**
   Kitten Fire, the Selma bust, Halemaʻumaʻu
6. **How it is built**
   detectors, the localization math, the learning loop
7. **Limitations & future directions**
   the honest ledger
8. **Sources & links**
   everything cited, everything public

Backup slides with the full case files — falsification, recovery, localization, controller, holdover watch — follow the closing slide for Q&A.

THE PLATFORM

Sage: AI at the edge, continent-wide

- An NSF (National Science Foundation)-funded cyberinfrastructure of AI-capable sensor nodes: the continental US, Alaska, Hawaiʻi, and Puerto Rico.
- 294 registered nodes — the 212 with public GPS (Global Positioning System) coordinates are mapped here, live from the Sage manifest.
- Each Wild Sage node carries cameras (fixed + steerable), a microphone, a weather station, LoRaWAN (long-range wide-area network) radio, and an on-node GPU (graphics processing unit).
- The Waggle software stack runs containerized machine learning at the edge; results flow to the Beehive cloud behind open data APIs (application programming interfaces).
- **Sage FlashPoint uses this fleet as-is: its cameras, microphones, and weather stations become the lightning instrument.**



Wild Sage node W0A4 at Argonne (photo: Sage / University of Hawaiʻi EPSCoR)

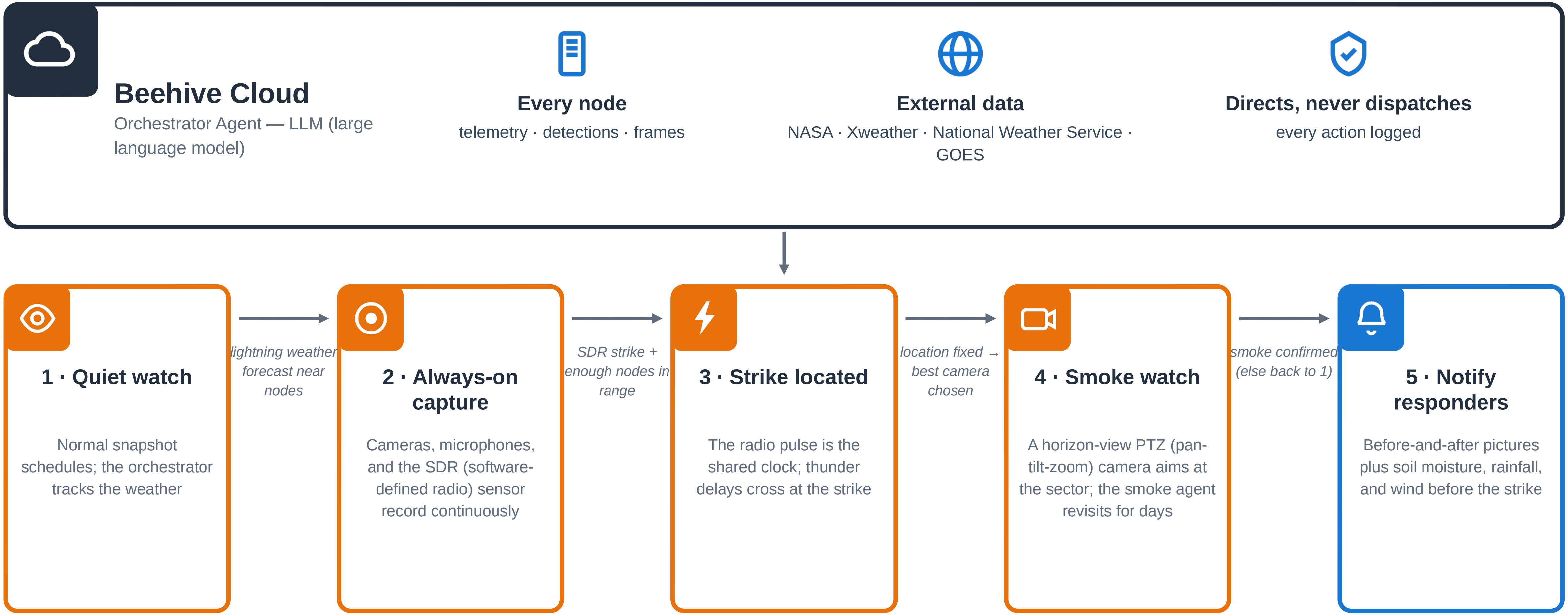
Hanwha PTZ (pan-tilt-zoom) camera — the steerable class on Sage nodes (image: Hanwha Vision)

Node locations: Sage manifest API · Basemap: Natural Earth

SAGE FLASHPOINT IN ONE PICTURE

Sage FlashPoint — flow of state

The Beehive-level orchestrator supervises every step; each arrow fires only when its condition is met.



SDR lightning sensing is planned hardware — the camera flash serves as the start-of-clock today. Notifications are human-reviewed before anything reaches responders. Soil moisture comes from NASA satellite and model data; an on-node probe is the planned upgrade.

WHY IT MATTERS

The strike is the earliest possible warning

- Lightning holdover fires smolder for hours or days before flaring. The strike location is the earliest warning that exists — and today nothing connects a strike to a follow-up watch.
- The Kitten Fire ignition storm played out six kilometers from a recording Sage node — while the node's steerable camera pointed at a cabin wall.
- Responders get a verified, evidence-backed watchpoint, not another false-alarm feed: detections nominate, satellites and cross-node agreement confirm, humans review.

6 km
node to fire point

145 min
storm-mode lead time

0/17
audio-only claims that survived

WHAT IT CONTRIBUTES

Detection with sensors already deployed
cameras + microphones + weather stations the fleet has today

An honesty architecture
0/17 falsification made nominate-only the design law — no sensor confirms alone

Tasking that closes the cabin-wall gap
storm mode armed 145 minutes early on the real-storm replay

Findings returned to Sage
soil-moisture gap, manifest ≠ archive, microphone seasonality, camera-siting recommendations

Transferable by design
the same controller re-armed on wind triggers is the Hawaiʻi variant (HCDP mesonet feeds)

THE DATA

Every claim traces to a named source

Sage node archives

W06C (Kitten): 779 audio clips, 2,739 PTZ frames, 51.7k weather records over July 1–4 2025 · W067 (Selma): 216 images, 25.9k weather records · 518 clips hand-classified for the retrospective.

Sage data APIs + MCP server

Public query API for telemetry; the Sage Grande MCP (Model Context Protocol) server's authenticated proxy carried every archive file download; its job tools are the planned path for live storm-mode control.

NOAA GOES GLM lightning

GOES-18 + GOES-19 Geostationary Lightning Mappers on AWS Open Data: the dual-satellite falsification, the 219-flash ignition storm, and a Jetstream2 bulk scan — 7.05M granules → 353 fires near nodes → 448 ranked candidates.

NASA Earthdata

POWER (Prediction of Worldwide Energy Resources) dryness at both fire sites; SMAP (Soil Moisture Active Passive) L3 soil-moisture retrievals via Earthdata Login — the moisture inputs the fleet itself cannot provide.

NIFC WFIGS fire ledger

The interagency incident record (IRWIN-fed): Kitten, Signal Flat, and the 14-fire Selma day, each with agency-determined natural cause. News (Buckrail) used as corroboration only, never as a case source.

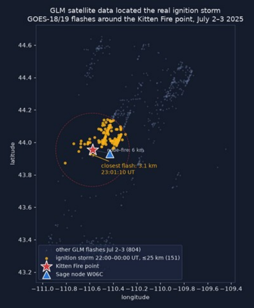
Live weather feeds

NWS (National Weather Service) forecasts and Vaisala Xweather lightning / storm-cell / threat endpoints — live-verified during camp week; these are the storm-mode controller's arming inputs.

All numeric claims in this deck derive from these six sources; the repository pins the exact queries and timestamps.

CASES OBSERVED

Three fires, three lessons — all from real archives



Kitten Fire — Grand Teton, Wyoming

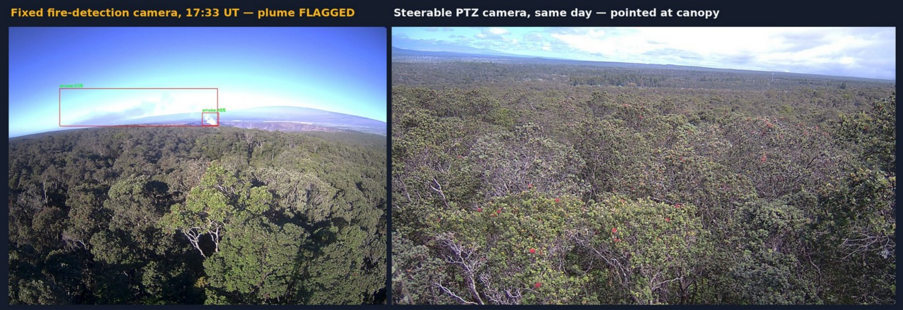
Discovered July 3 2025, six kilometers from node W06C, natural cause. A dry night storm (0.0 mm rain) two nights before; the satellite scan put the closest flash 3.1 km from the fire point. 22 thunder arrivals recovered from the node's own audio archive. Signal Flat (7.7 acres) hit the same node three weeks later.

Lesson: the evidence was all there — nothing was tasked to look.

The Selma bust — southwest Oregon

Fourteen natural-cause fires logged in ONE day (July 8 2025), 14–34 km from node W067. The node's own gauge: 0.06 mm of rain across five days under a 37 °C spike — its only trace lands exactly on the fire line.

Lesson: met data alone already separates dangerous storms — risk scores 87/84 vs 41 for the wet control.



Halemaʻumaʻu plume — Hawaiʻi Volcanoes NP

Node W097's fixed fire-watch camera flagged the persistent gas plume with its production detector — the same day its steerable PTZ camera studied canopy.

Lesson: camera capability was never the gap; watch tasking is.

Images: GLM storm scan (NOAA via AWS), W067 gauge trace vs the WFIGS fire line, W097 camera pair (production detector's boxes).

HOW IT IS BUILT

The pipeline — and the math under it

LOCALIZATION MATH

$c = 343 + 0.6\,(T - 20\ °C)\ \ m/s$

speed of sound from the node's own thermometer

$r_i = (t_{thunder,i} - t_{flash}) \times c$

flash-to-bang range: light is the shared time-zero — no cross-node clock sync

$\hat{p} = \operatorname{argmin} \Sigma\ [\ (|p - a_i| - r_i)\ /\ \sigma_i\ ]^2$

trilateration: weighted least squares over range circles — Gauss–Newton, multistarted from circle-pair intersections

consistency: $\leq$ 4,096 range combinations

RANSAC-style sweep, one candidate per node; lowest joint residual wins — rain artifacts don't agree across nodes, real thunder does

gate: $GDOP = \sqrt{tr(\ (J^TWJ)^{-1}\ )} \leq 30$

poor geometry is downgraded to 'ambiguous', never over-claimed; the 1-σ ellipse comes from the same covariance

fallback: thunder-only TDOA

time-difference-of-arrival multilateration; Monte Carlo on real node coordinates: 135–331 m median error

Edge detectors (DSP — digital signal processing)

Flash = scene-luminance spike; thunder = noise-adaptive, spectrally whitened onset (leading edge, not loudest peak); anchored re-listening around satellite flash times.

Classifier probe v0 — the first neural result

Frozen YAMNet audio embeddings + logistic regression, 5-fold cross-validation: AUC (area under the curve) 0.952 vs 0.699 for the DSP score. Reviewer clicks accumulate as the next label set.

Storm-mode controller

IDLE → OUTLOOK → APPROACH → STORM → AFTERMATH; budget caps, hysteresis, audit log. 19/19 tests; armed 145 minutes early on the real-storm replay.

Agent skillset on camp node H03E

Smoke-watch skills served through the real sage-agent gateway: Gemma 4 captions frames, YOLO11 draws boxes; cloud escalation via NVIDIA NIM is restricted to text after a pixel-hallucination check.

Jetstream2 fleet-history catalog

GLM bulk scan on ACCESS Jetstream2: 7.05M satellite granules; five years of storms near nodes ranked into 448 retrospective candidates — Kitten lands #4.

SDR (software-defined radio) — planned

The sferic radio pulse as a daytime time-zero, replacing the camera flash. No SDR is deployed on the fleet today; this is the designed third modality.

THE HONEST LEDGER

Limitations — and exactly what comes next

LIMITATIONS

- One real storm validates the audio chain (Kitten); fusion + flash are validated in replay tests — real multi-node captures need storm mode running live.
- Ground truth (satellite pixels 8–14 km) is coarser than the sub-km claims — Vaisala research-data request submitted July 24, not yet approved.
- Read access to key fire-country node archives (W067, W084/W06F, W019) was requested but not granted — additional storm/fire cases went unexamined.
- Live PTZ control untested: the W0A4 sandbox camera's web interface was reached over an SSH (Secure Shell) tunnel, but control permissions were not granted — re-aim runs in test mode only.
- No smoke model runs at the edge yet — SmokeyNet is adopted and planned, detection boxes today come from YOLO11.
- One hackathon week: v0.1 risk weights, dry-run job control.

FUTURE DIRECTIONS

- **NLDN (National Lightning Detection Network) ground truth** research request submitted July 24
- **Neural classifier** probe v0 landed: AUC 0.95 vs 0.70 DSP — next, reviewer-click labels
- **Smoke leg** SmokeyNet plugin on watch cadence, horizon-band + dwell rules
- **RF (radio frequency) time-zero** SDR sferic anchor — daytime parity with the night-strong camera flash
- **Hawaiʻi variant** wind-trigger controller on HCDP (Hawaiʻi Climate Data Portal) mesonet feeds
- **Live PTZ actuation** camera-control credentials on the W0A4 sandbox unblock real re-aim
- **Beehive orchestrator agent** instantiate the fleet-level agent the skillset was shaped for (sage-agent)

SOURCES

Everything cited — everything public

DATA

- NIFC WFIGS Incident Locations — fire records, agency-determined cause
- NOAA GOES-18/19 GLM — AWS Open Data registry
- NASA POWER — dryness (GWETTOP) at both fire sites
- NASA SMAP L3 soil moisture — via Earthdata Login (CMR + EDL)
- NWS API (api.weather.gov) — forecasts & outlooks
- Vaisala Xweather — lightning, storm cells, threats (live-verified)
- Sage data & storage APIs; Sage Grande MCP server (mcp.sagecontinuum.org)
- Buckrail (July 3 2025) — news corroboration only

PRIOR WORK

- Sage / Waggle testbed — nodes, APIs, audio-sampler, ptz-yolo, smoke-detector plugins
- sage-smoke-detection — official SmokeyNet plugin (HPWREN/FIgLib-trained)
- sage-agent — the agentic gateway the smoke-watch skills target
- Sage Autonomous Camera Control (Dematties et al.) — PTZ plumbing
- Sage SDR lightning project — the planned RF third modality
- NDP Sage Smoke Detection Workflow (I. Perez, SDSC) — preprocessing recipe

MODELS & SOFTWARE

- YAMNet (Google Research) — frozen audio embeddings
- Ultralytics YOLO11 — detection boxes in gateway runs
- Gemma 4 via Ollama — vision captioning at the edge
- GLM-4.x via NVIDIA NIM — text-only cloud escalation rung
- Leaflet — maps © OpenStreetMap contributors, © CARTO
- sage_data_client, pywaggle, NumPy/SciPy, soundfile, Streamlit + Altair

Code, data, audio & writeup: **github.com/scwatson4/sage-summer-camp-2026 → flashpoint/**
Edge plugin on the Sage app portal: **portal.sagecontinuum.org/apps/app/scwatson/flashpoint-thunder**

Thank you

The fleet already hears the storms — Sage FlashPoint makes it listen.

SW

Samuel Watson

GRADUATE RESEARCH ASSISTANT

University of Hawaiʻi at Mānoa

Email **scwatson@hawaii.edu**

LinkedIn **linkedin.com/in/swatson71432**

SCAN TO CONNECT ON LINKEDIN

linkedin.com/in/swatson71432

BACKUP — TARGET OPERATING CONCEPT

Flow of state: from quiet skies to a verified alert

Beehive cloud — orchestrator agent

A more capable LLM (large language model) agent runs where Sage already connects every node — it moves nodes between states; the nodes stay deterministic.

Sees every node
telemetry, detections, camera frames, health — via the Beehive pipeline

Sees the outside world
NASA Earthdata · Vaisala Xweather · National Weather Service · GOES satellites

Directs, never dispatches
chooses node states and cameras; every action logged

The orchestrator supervises every state at right and can advance or reset the flow at any time.

1 — Quiet watch
Normal snapshot schedules; the orchestrator tracks forecasts and radar for every node's neighborhood.

↓ *weather conditions suggest lightning near a group of nodes*

2 — Always-on capture
Cameras, microphones, and the SDR (software-defined radio) lightning sensor record continuously on the affected nodes.

↓ *SDR registers a positive strike AND enough nodes sit within listening range*

3 — Strike triangulated
The strike's radio pulse gives every node the same start-of-clock; thunder delays give distances that cross at the strike.

↓ *location fixed — the orchestrator picks the best-placed camera*

4 — Smoke watch
A PTZ (pan-tilt-zoom) camera with a horizon view aims at the strike sector; the smoke agent revisits through the holdover window.

↓ *smoke confirmed at the sector · no smoke by window's end → return to 1*

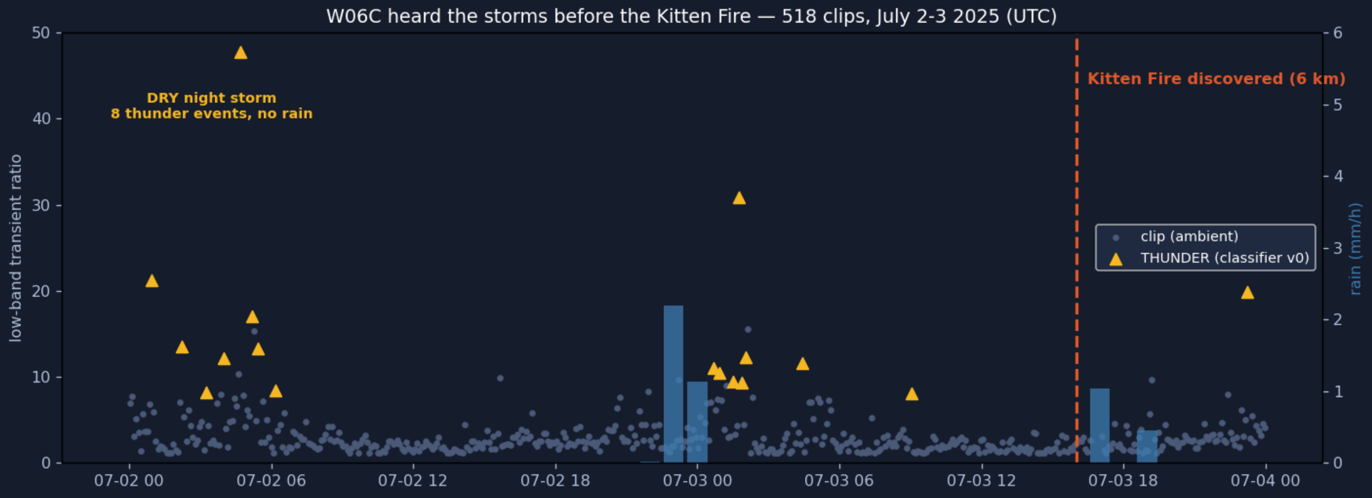
5 — Notify emergency personnel
Before-and-after pictures plus the conditions leading up to the strike — soil moisture, rainfall, wind.

SDR sensing is planned hardware — the camera flash is the start-of-clock today. Notifications are human-reviewed before anything reaches responders. Soil moisture currently comes from NASA data; an on-node probe is the planned upgrade.

BACKUP — THE CASE

A Sage node heard the fire start. Nobody was listening.

- Kitten Fire — Grand Teton WY, natural cause, discovered midday July 3 2025, six kilometers from node W06C.
- The node recorded straight through both suspect storms: audio snapshots, PTZ (pan-tilt-zoom) camera frames, full weather telemetry.
- Holdover fires smolder for hours or days before flaring. The strike is the earliest possible warning — and today nothing connects strikes to a follow-up watch.



Across 518 classified clips: a DRY night storm (0.0 mm rain) and a second storm the next evening — then the discovery line.

6 km
node to fire point

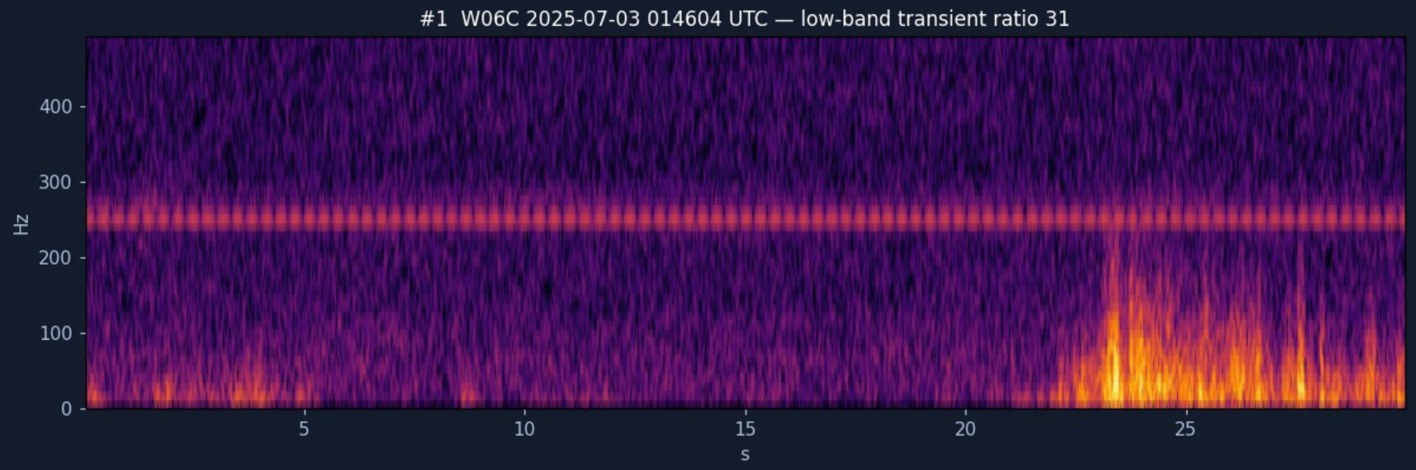
779
audio clips, July 1–4

51k
weather records

17 confident detections. Two satellites. Zero survivors.

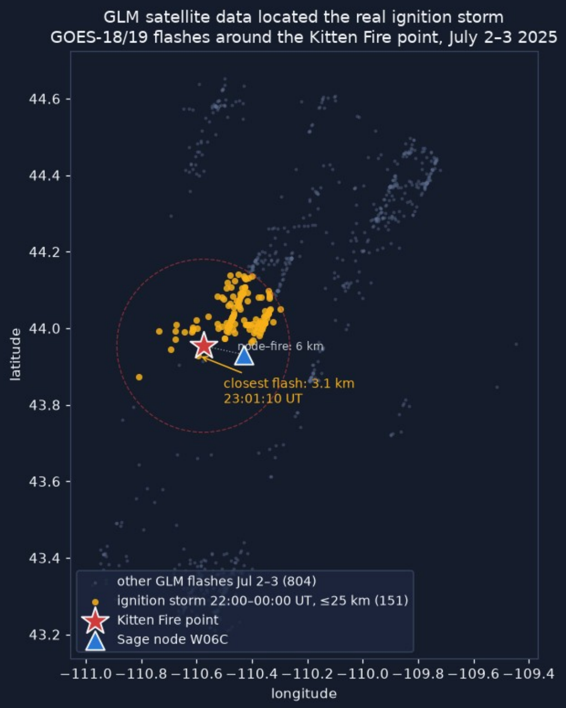
0 / 17

- Audio-only detector v0 flagged 17 thunder events across the storm nights — all confident, none wind-vetoed.
- Cross-checked against BOTH GOES-18 and GOES-19 (Geostationary Operational Environmental Satellites) lightning sensors — ±90-second window: not one flash within 50 km of any of them.
- **Single-modality detection false-alarms confidently.**
- That result became the architecture: standalone detections are nominations — never confirmations.



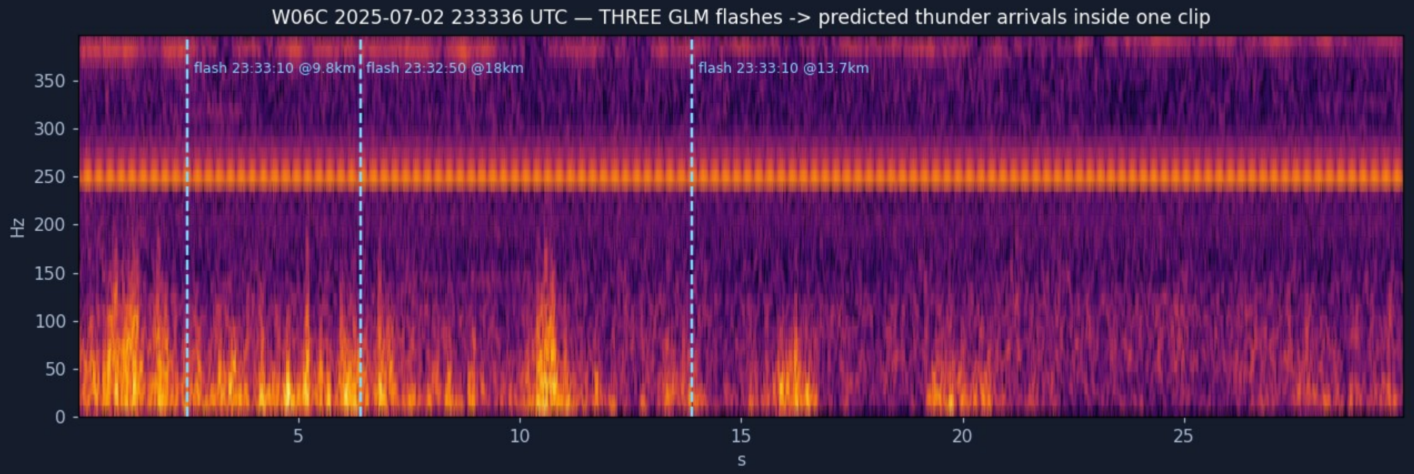
The strongest candidate of the 17 — low-band transient ratio 31 — of undetermined origin per two independent satellites.

Anchor on the flash, and the buried thunder appears



The real ignition storm, located by satellite scan: closest flash 3.1 km from the fire point, ~3 h before discovery. Data: NOAA (National Oceanic and Atmospheric Administration) GLM (Geostationary Lightning Mapper) via AWS (Amazon Web Services) Open Data.

- Each satellite flash gives a time-zero. Sound needs ~3 s per kilometer — so re-listening is confined to each flash's predicted arrival window.
- 22 real flash → bang arrivals surfaced from clips the rain noise had blinded — including three flashes ranged in a single 30-second clip:



22
arrivals recovered

20/22
detector recall

0.7 km
median range error

BACKUP — LOCALIZATION

Range without clocks, fixes without over-claiming

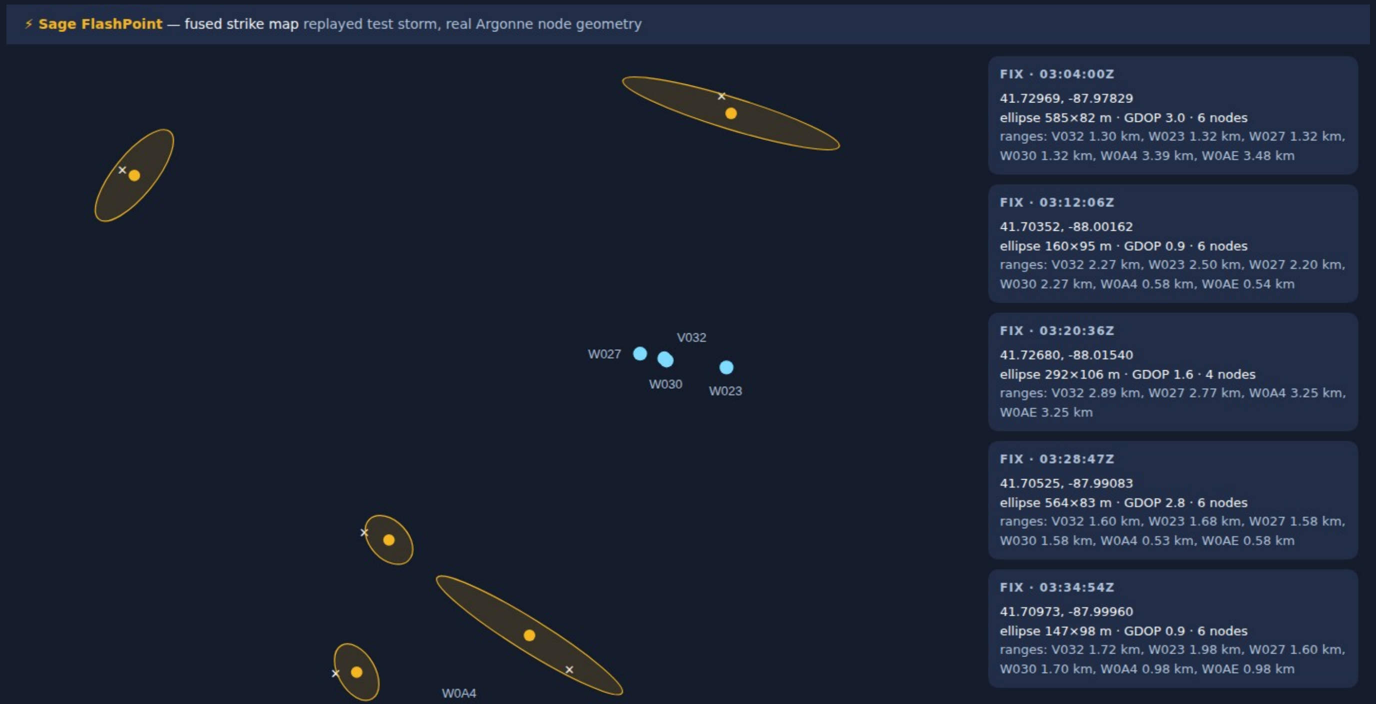
- Light arrives instantly — a camera flash is a free, absolute time-zero at every node. No cross-node clock sync needed.
- range = Δt × 343 m/s, per node; several nodes' ranges triangulate the strike.
- Cross-node consistency (RANSAC — random sample consensus) picks the one candidate set that agrees; a geometry gate labels every solution fix, ambiguous, or range-only — the system never over-claims.

96 m

fix with no satellite, no sync (integration test)

5/5

strikes fixed in the replayed test storm, 70–273 m



Fused strike map: a replayed test storm over the real Argonne node geometry (cameras + microphones only).

BACKUP — THE CONTROLLER

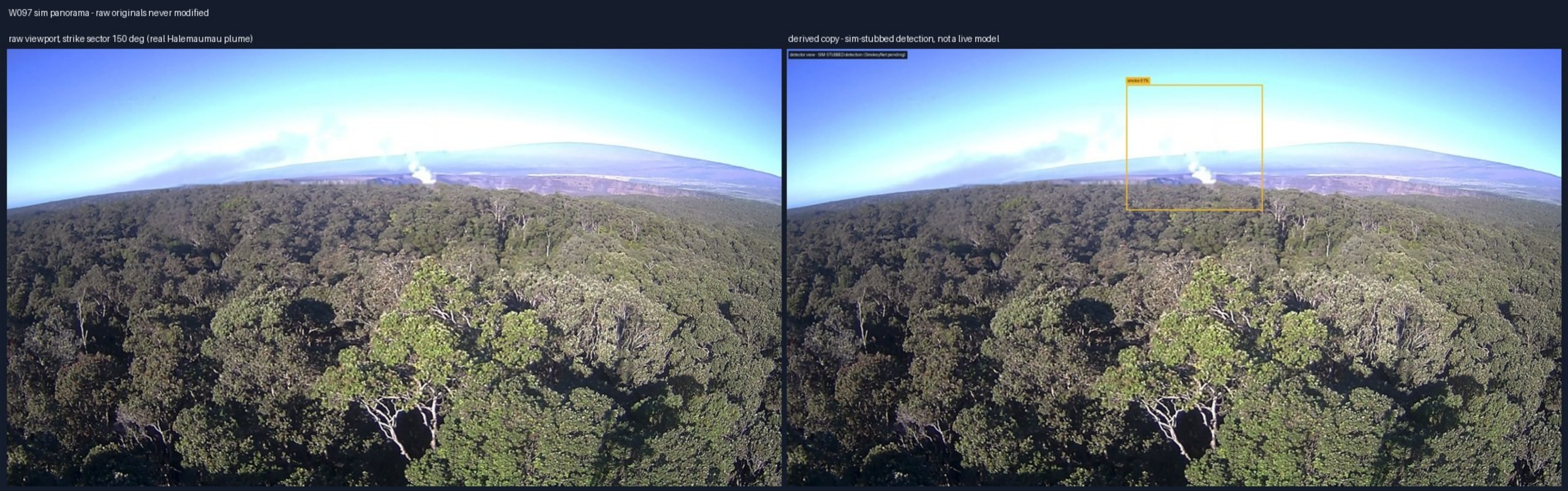
Storm mode armed 145 minutes before the first flash

IDLE → OUTLOOK → APPROACH → STORM → AFTERMATH

- Free NWS (National Weather Service) outlooks arm it cheaply; radar cell tracks and local sensors escalate it; every action is budget-capped, hysteresis-damped, and audit-logged.
- **Replayed against the real Kitten storm feeds, it armed 145 minutes before the first local flash — then scheduled the holdover watch and expired cleanly. 19/19 tests.**
- Why it matters → the frame on the right: through the entire ignition storm, the node's steerable camera pointed at a cabin wall.
- Two failures, honestly separated: tasking (fixed by this controller) and siting — W06C sits low in forest, so smoke confirmation needs elevated partner cameras or better-sited nodes. Both remedies are recommended.

W06C's PTZ camera during the ignition storm (top rows) and the falsified events (bottom row).

Every strike becomes a 72-hour watchpoint



Evidence ships as an untouched raw frame plus a labeled derived copy. The boxed detection here is a test-harness placeholder — in live gateway runs, Gemma 4 captions each frame and YOLO11 draws detection boxes; SmokeyNet is the planned smoke detector (W097 panorama; the plume is a real Halemaʻumaʻu gas plume).

- PTZ patrols strike sectors by risk, dwelling to the smoke model's needs — validated through the real sage-agent gateway on the camp node (test-mode PTZ, live Gemma 4 captions).
- Ignition-risk score with per-factor provenance: node rain gauge, NASA (National Aeronautics and Space Administration) POWER (Prediction of Worldwide Energy Resources) dryness, fuel, detection confidence.
- Alerts land in Slack: one glanceable summary line, evidence strip inline, full forensic audit in-thread.
- Reviewers click Confirm / False positive / Keep watching — decisions flow back to the node AND become labeled training data. No automated dispatch, ever.

BACKUP — THE SECOND CASE

Fourteen fires. Five days. 0.06 millimeters of rain.

Queried live during camp week from W067's public weather streams. Fire records: NIFC (National Interagency Fire Center) WFIGS (Wildland Fire Interagency Geospatial Services), agency-determined cause.

- Selma, Oregon: 14 natural-cause fires logged in ONE day, 14–34 km from node W067.
- The node's own gauge: 0.06 mm across five days, under a 37 °C spike. Its only movement — a 0.05 mm trace — lands exactly on the fire line: the ignition storm itself, essentially dry.
- **Risk score v0.1 back-test — separation is already clean:**

87	Kitten dry night storm → fire next day
84	Selma bust → 14 fires next day
41	wet-storm control → low risk, correct
0	no-strike day → no card, correct

BACKUP — FOR Q&A

The pattern repeats on a second node

Fixed fire-detection camera, 17:33 UT — plume FLAGGED

Steerable PTZ camera, same day — pointed at canopy

W097, Hawaiʻi Volcanoes National Park, 2025-10-05. Left: the node's fixed fire-detection camera flagged the persistent Halemaʻumaʻu gas plume — the detection boxes were drawn by the node's production detector, not by this project. Right: the steerable PTZ camera the same day, pointed at canopy.

Camera capability was never the gap — watch tasking is. The same lesson as the cabin wall, on a different node and hazard.